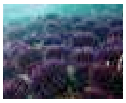
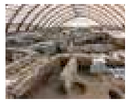


Kelp forests responding to urchins
Urchins are taking over kelp forests along the northern California coast.
http://dgit.in/apr21-43



Ancient city of Çatalhöyük
A dig in Turkey has revealed the ancient city of Çatalhöyük.
http://dgit.in/apr21-44

Fibre seismology

Existing optical fibres are being used by geophysicists to detect minute movements in the ground and seas with incredible resolution

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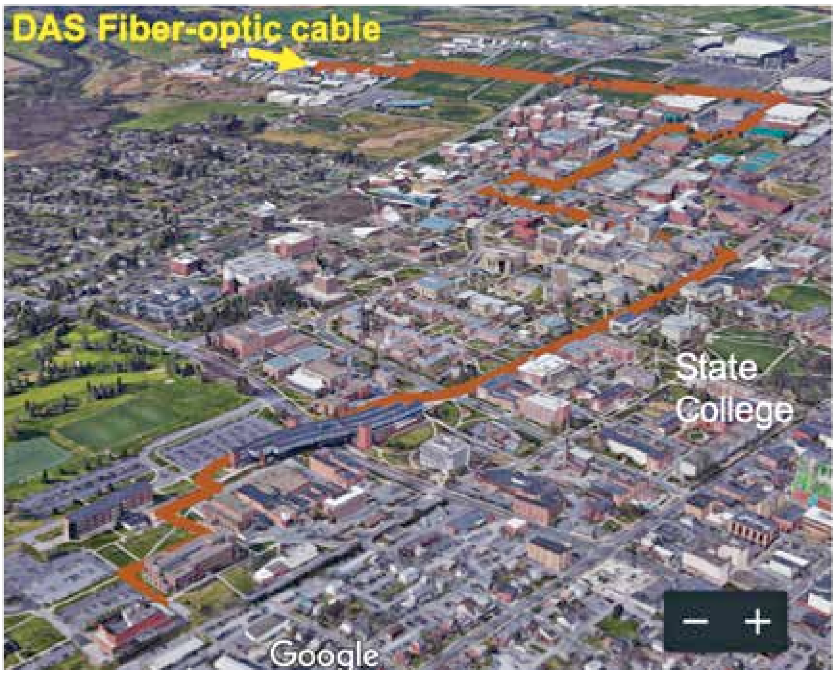
During the dot com bubble there was tremendous speculation on the future prospects of the technology industry, and one of the fallouts was that telecommunications companies invested heavily in fibre optic cables. This problem was exacerbated by the fact that once the ducts were in place, laying additional fibre lines did not take up much extra costs. At around the same time as the cables were being put in place, telecommunications engineers improved the rate at which data could be transferred over a single cable, and the capacity rose by a factor of a 100 after different wavelengths of lasers began to be used to relay data across a single fibre. This led to a buildup of a lot of spare capacity in unused fibre, known as "dark fibre". The fibre that is not being used actively for communication is known as dark fibre, while the fibre that is being used is "lit". There have been many studies in and around California that use existing communications infrastructure for geosensing. The Oil and Gas industry developed and has long employed the principle of distributed acoustic sensing (DAS) to detect leaks and cable faults. The technology has also been used to monitor seismic activity in Alaska, Iceland and California.



Credit: Penn State

The FORSEE network used by researchers at Penn State

Traditionally, the devices used to monitor seismic activity are known as geophones, but these are expensive to deploy and maintain, especially in densely populated urban areas, and as such cannot be relied on to provide long term data. For undersea monitoring, the devices are known as Ocean-bottom seismometer (OBS), but permanent arrays of these are expensive to install as well. However, existing fibre optic networks that act as geophones because of distributed acoustic sensing reduces the cost and time required to set up geosensing networks.

In 2018, a team of researchers reported on successfully using lengths of cable in the UK and other European countries to detect a number of earthquakes. The data was high resolution, and provided information on the seismic activity along the entire length of the cable. Google Engineers took note of the study, and started thinking of how they could use their undersea cable networks for geosensing applications, or help such researchers in their studies. In 2019, a team of researchers used a dark fibre testbed in Sacramento, California, to monitor land activity. The cables work on the principle of measuring the changes to the characteristics of light, as it passes through the stress and strains experienced by the microscopic fibres. Essentially, a virtual motion sensor is created along every two meters of the cable, so a stretch of 20 kilometers has 10,000 sensors. The team went on to monitor deep sea seismic activity in an undersea cable that stretched to a seismic activity station in the Pacific Ocean, that started off in Montreal Bay, California. Over four days, they were able to detect a 3.5 magnitude earthquake, as well as the sound of ocean waves.

Also in 2019, Google Engineers began to monitor the spectral characteristics of their global submarine cables, and on January 28, 2020,